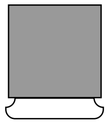
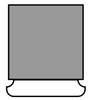
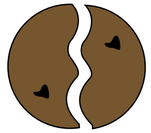
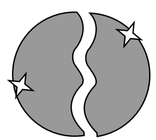
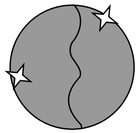
Object: Trigger

Broken	Whole
	

Properties	• Binary toggle state, Broken or Whole • Accessability clarity, Broken has defined cross design and stone permutation patterns whereas Whole is smooth-faced
Behaviours	• Toggle switches between states on any instance of contact/force
Game State Feedback	• Audio ▪ Sound effect ◦ Plays when activated • Visual ▪ Colour ◦ Changes based on state ▪ Pattern ◦ Changes based on state
Systemic Feedback	• Toggle event triggers other objects

Toggle: Stone

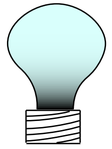
Stone \ Toggle	Broken	Whole
		
Non-Conductive		
Conductive		

Object: Lightbulb

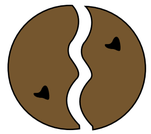
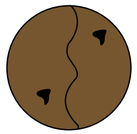
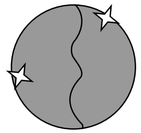
Permutations	Unpowered	Powered
Whole		
Broken		

Properties	• Binary Powered/Unpowered (Trigger) states and Whole/Broken states (Relationship) • Accessability Pattern ▪ Unpowered is blank, powered has lightning bolt symbol
Behaviours	• Bulbs power and depower in response to trigger object • Unable to be split via trigger but can be restored Whole if nothing is occupying the space... • Vulnerable to high crushing force... • Toggle the state of something else when all bulbs are powered or depowered
Game State Feedback	• Audio ▪ Crackling of electricity sound plays when powered ▪ Fizzle out of electricity sound plays when depowered ▪ Shattering glass sound upon becoming broken ▪ Ceramic tinkering noise when becoming whole • Visual ▪ Animation ◦ Electricity moving and flowing in response to Trigger ◦ Small Glass chip explosion when breaking, reversed when becoming whole ▪ Shape ◦ Whole is smooth and clean, broken is rough and jagged ◦ Powered is denoted by extra symbols
Systemic Feedback	• Trigger a toggle event when all bulbs are powered or unpowered

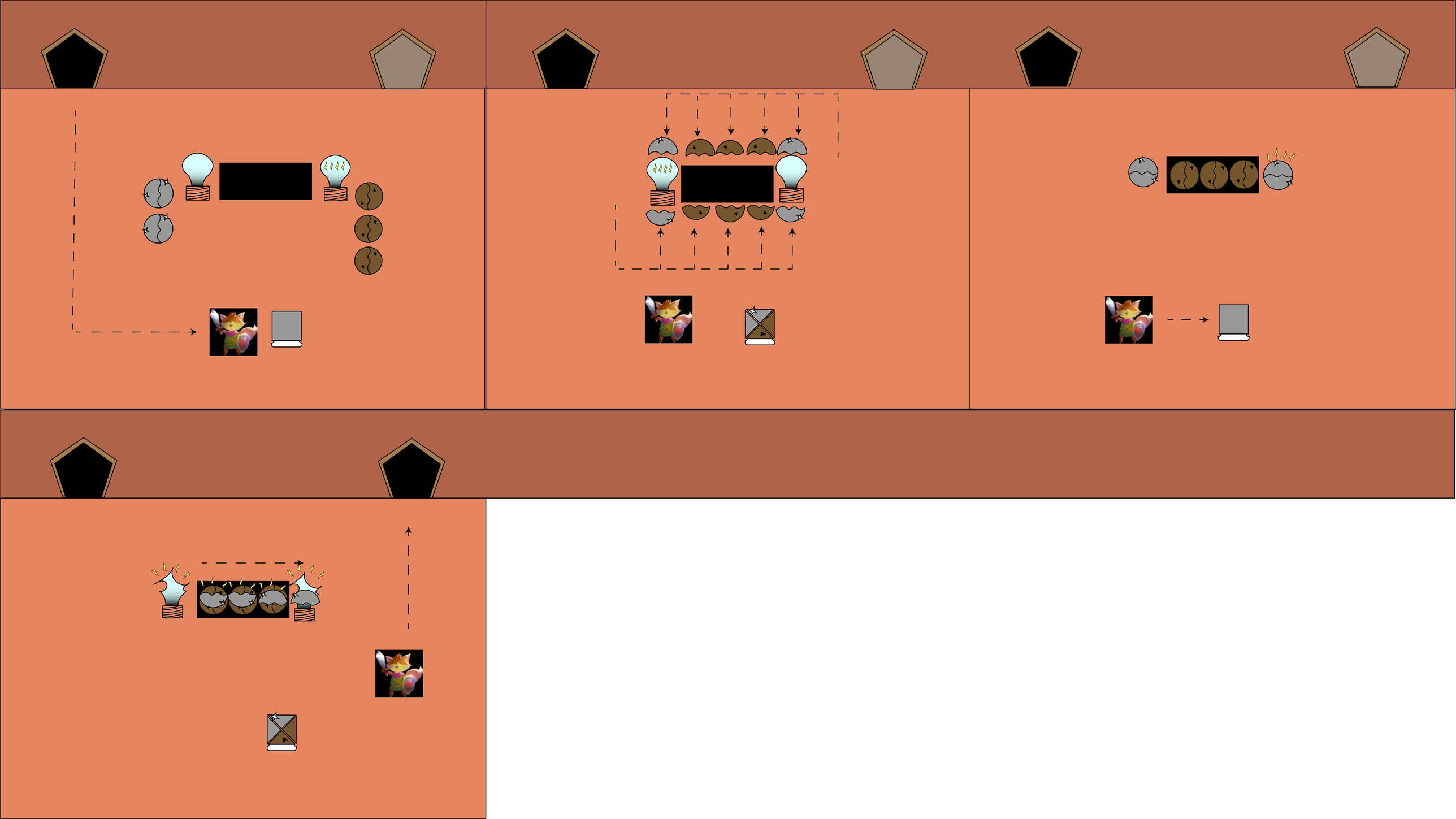
Toggle: Lightbulbs

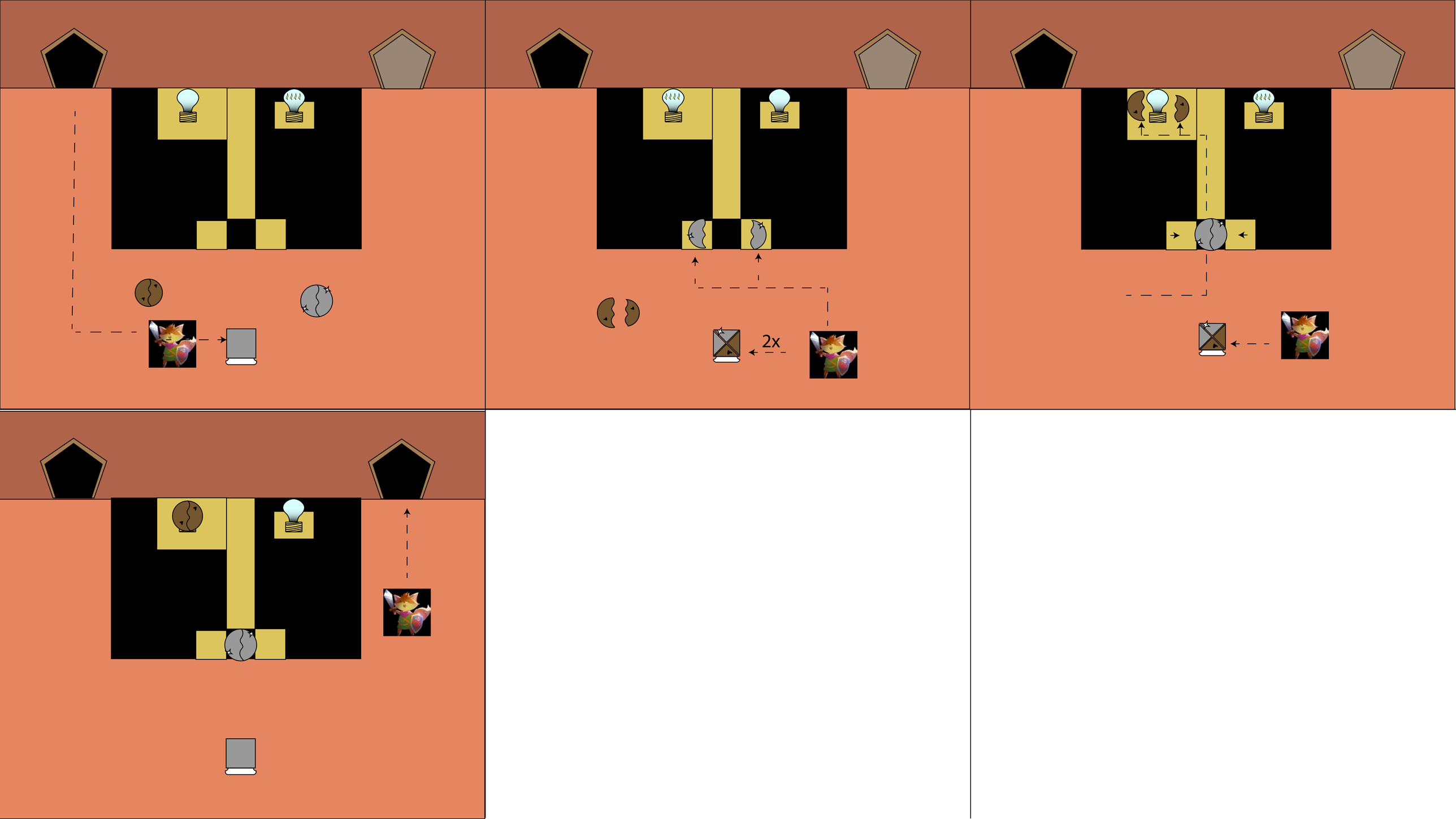
Lightbulbs \ Toggle	Broken	Whole
Powered		
Unpowered		

Object: Stone

Permutation	Broken	Whole
Non-Conductive		
Conductive		

Properties	• Binary broken or whole states based on toggle switch • Colour ▪ Brown or Silver dependent on type/behaviour • Accessibility pattern ▪ Conductive has 4-point stars, Non-Conductive has black spots
Behaviours	• Conductive or Nonconductive based on colour • Occupy space; ▪ Blocking projectiles ▪ Blocking player from passing through directly ▪ Lays on topmost • When Broken ▪ Halves can be moved individually ▪ Can only move along the X and Y axis, and will slide to remend to closest other half when triggered ▪ Can be rotated/turned ▪ Take up only half a tile width-wise... • When Whole ▪ Too heavy to be moved by player ▪ Full/Big enough to fill holes in the ground ▪ Wide enough to be walked on top of from above
Game State Feedback	• Audio ▪ SFX ◦ Crumbling or Reassembling Sound when split or mended ◦ Faint twinkling noise when near conductive stones • Visual ▪ VFX ◦ Debris falling off when breaking stones ◦ Slow rotation of stars on Conductive Stones ◦ Non-Conductive stones smothering a bolt or two when blocking a current/charge ◦ Conductive stones radiating electricity when carrying a current/charge ▪ Shape ◦ Visual changes when Broken vs Whole
Systemic Feedback	• Can walk on top of Whole Stones • Can't pass through Whole Stones • Non-Conductive block any power from coming through when occupying a space with it • Conductive stones carry a charge even when broken and halved




2x

Assignment 2:
Systemic Games with Emergent Systems

Course: GAME 10017 Game Systems Design
Professor: Raphaël Tétreault
Designer: Elijah-kole Francis
Date: Feb 19, 2026

Assignment 2:
Systemic Games with Emergent Systems

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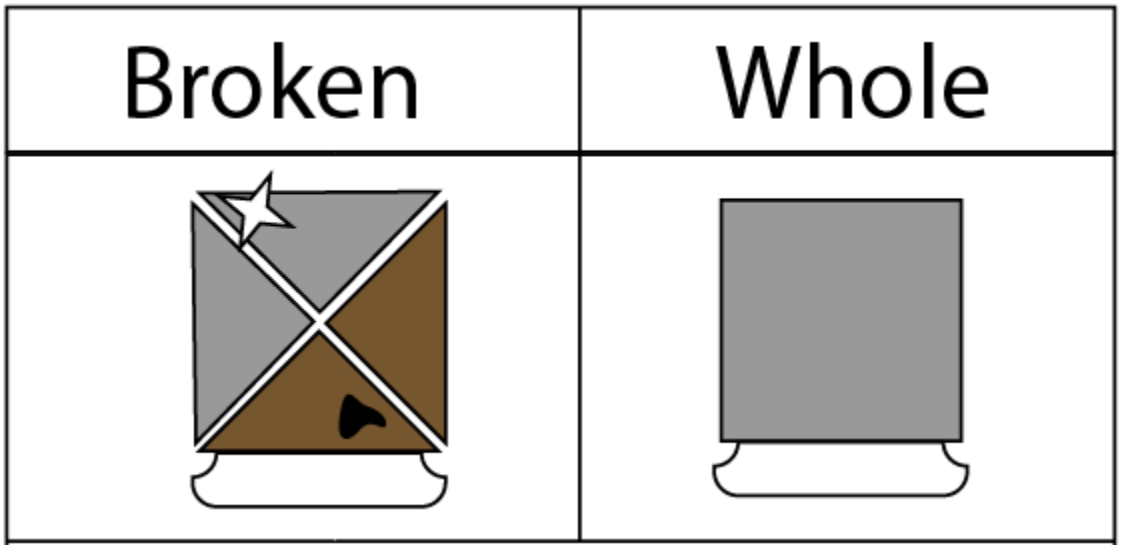
1.0 Assumptions

The player is assumed to have basic top-down classic zelda tools such as a close range sword, and long range tools like arrows and bombs but no overspecific items or abilities that would tweak the entire experience to be based around them. (Such as the Wall Merge mechanic from Link Between worlds for example.) In addition, the player is also assumed to have the ability to grab, move, drag and push an item as well as the ability to warp out of a room or back to a dungeon entrance to prevent soft locks.

1.1 Trigger Switch

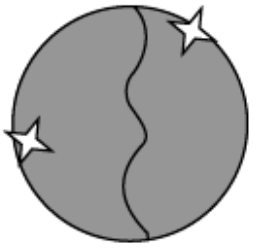


This object's design is to create direct coupled interactions between it and other objects within the game. Working as a binary state switcher, it switches between broken and whole states when activated and is a core property to interact with. In addition it will need to be activated whenever the player sees fit in order to solve puzzles in the level based on the game environment around them.

The object signals changes in state as events to the other listening objects, the Stones and the Lightbulbs as detailed further in the document. It's capable of creating context-dependent interactions due to not having behaviours of its own, instead only

exhibiting systemic feedback. Thus, based on the puzzle or room it can generate relationships.

1.2 Stone

Permutation	Broken	Whole
Non-Conductive		
Conductive		

This object's design is to change binary states and enable the player to act directly with it, breaking if it's whole and remending if it's broken, in relation to the Trigger it has entirely coupled interactions. It also has direct coupled interactions with the Lightbulbs being the only tool the player has capable of changing their state to the broken permutation. The utilization of the behaviours each permutation has in each state is what will allow the player to complete or progress through each puzzle space or room.

The abundance of behaviours each permutation of *this* object has made it capable of creating various context dependent interactions based on the layout of the surrounding room or level. The object must be utilized in tandem with the Trigger to make the most of them in each permutation. Thus it has the most coupled interactions but also the most potential for context dependent interactions in any given space due to how much the player can interact with it in terms of relationships.

1.3 Lightbulbs

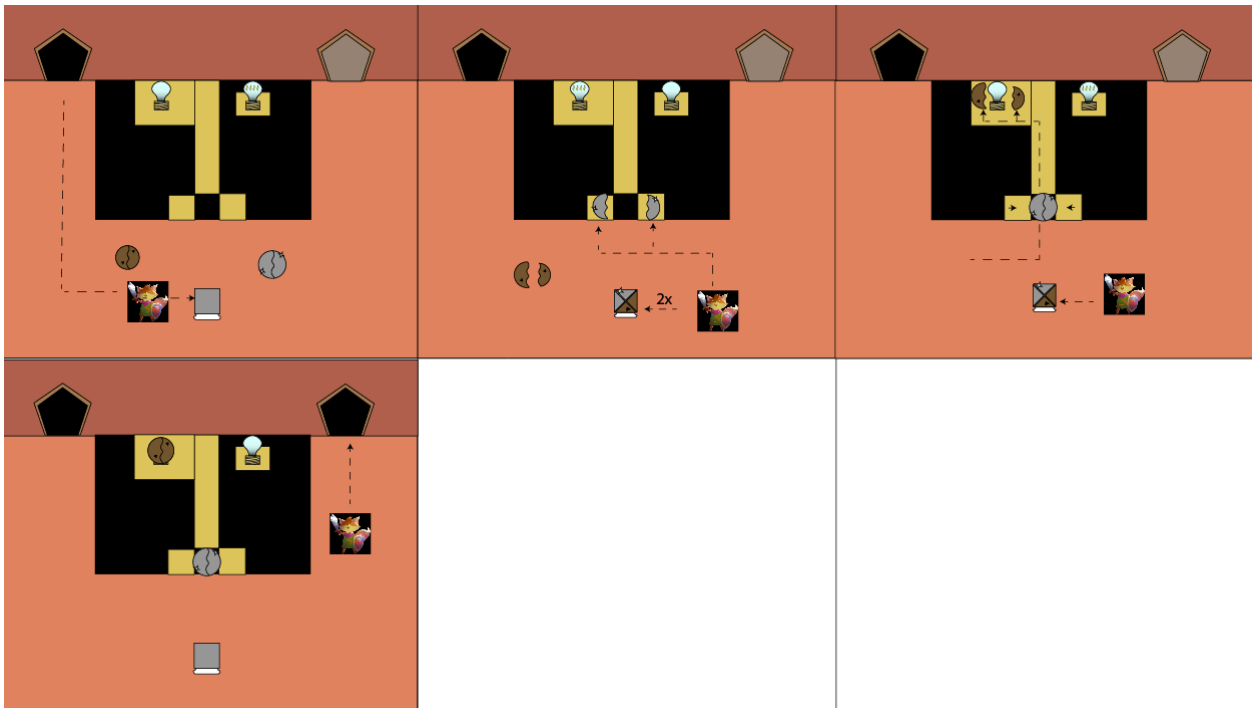
Permutations	Unpowered	Powered
Whole		
Broken		

This object's design serves as a signal or objective to the player as they aim to render all bulbs in a stage to the same powered or unpowered state. These objects also act in relation to the Trigger, powering and unpowering lightbulbs on each activation as a coupled interaction. They also can be made whole and re-mend with the trigger if they're broken, but not broken by that same trigger. Their coupled interaction with the Stones comes down to using them to break the bulb, and manipulating the powered or unpowered states with said stones as the player desires to solve the greater puzzle in the room. Due to this, no less than two lightbulbs can ever be present to create a puzzle, and they must be on alternate powered permutation states.

While the lightbulbs have several behaviours, only in relation to the other objects as coupled interactions, it is their systemic feedback of triggering something within the room or level that is the greatest value it serves. This however creates context dependent interactions as it relies much more on the current layout of the room or puzzle including elevation, number of bulbs, and which state will open the door that the player would engage with and complete.

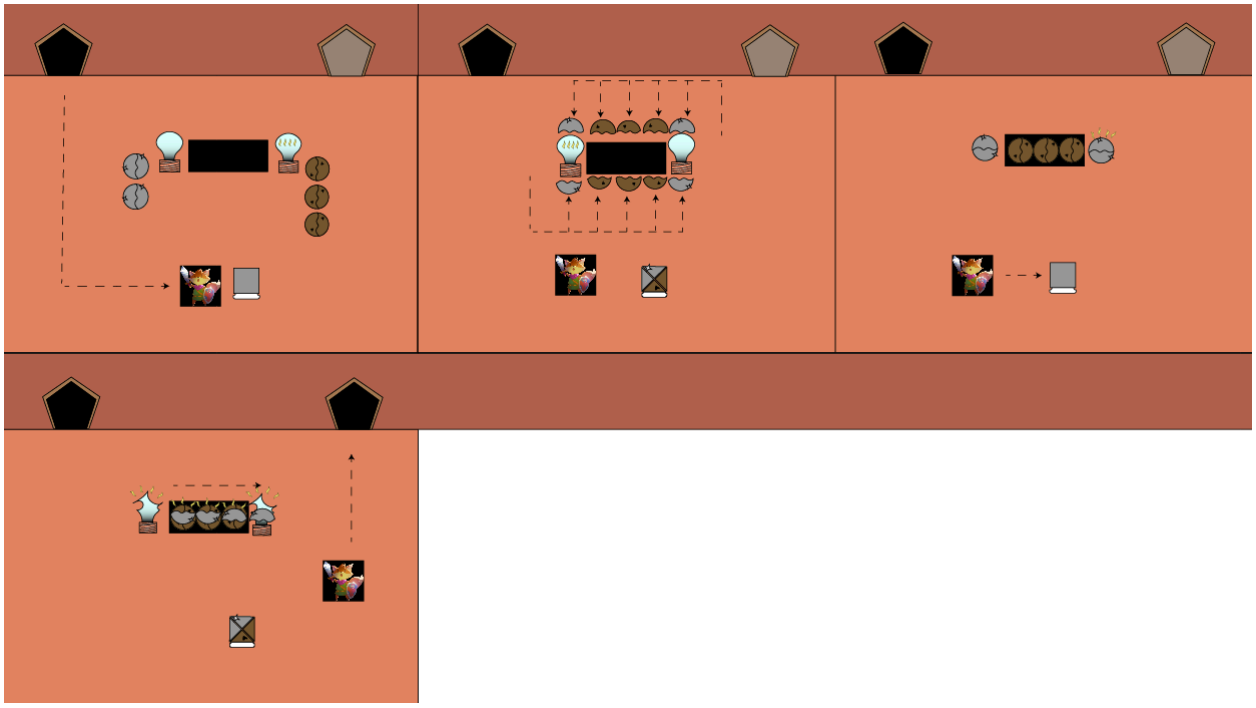
2.0 Scenarios and Possibility Space

(Figure 1. Repeated Trigger states, hole binding)



Within the first scenario, the player is given ample freedom and direction with two different types of stones and a small pathway to figure a way over and traverse. The complexity comes in with which stone to use to cross and at what point the trigger must be set to either Broken or Whole, and in this circumstance it must be engaged several times to succeed. It also shows that when full stones are confined to small spaced, even when broken, without space to go anywhere they'll act as whole. If the player uses the conductive stone to try and stem the power, it doesn't work as it would still flow through, whereas the non-conductive stone does stem the power and render both bulbs unpowered.

(Figure 2. Makeshift wiring)



In Figure two, the player is provided with a total of 5 stones, 3 non-conductive and 2 conductive. Once again like the first scenario there's a small complexity that comes with deciding which stone to use as a bridge but even beyond that, as whole stones the conductive stones wouldn't be long enough to link the powered and unpowered broken bulbs. Only when each half of the conductive stones is used as a makeshift wire, planted over top of the unpowered to let it flow and connect it directly adjacent to the powered dos it allow the player to complete the puzzle and proceed out of the level.